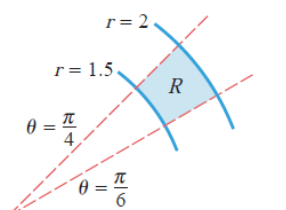


14.3 DOUBLE INTEGRALS IN POLAR COORDINATES

14.3.1 DEFINITION A *simple polar region* in a polar coordinate system is a region that is enclosed between two rays, $\theta = \alpha$ and $\theta = \beta$, and two continuous polar curves, $r = r_1(\theta)$ and $r = r_2(\theta)$, where the equations of the rays and the polar curves satisfy the following conditions:

- (i) $\alpha \leq \beta$ (ii) $\beta - \alpha \leq 2\pi$ (iii) $0 \leq r_1(\theta) \leq r_2(\theta)$

REMARK Conditions (i) and (ii) together imply that the ray $\theta = \beta$ can be obtained by rotating the ray $\theta = \alpha$ counterclockwise through an angle that is at most 2π radians. This is consistent with Figure 14.3.2. Condition (iii) implies that the boundary curves $r = r_1(\theta)$ and $r = r_2(\theta)$ can touch but cannot actually cross over one another (why?). Thus, in keeping with Figure 14.3.2, it is appropriate to describe $r = r_1(\theta)$ as the *inner boundary* of the region and $r = r_2(\theta)$ as the *outer boundary*.



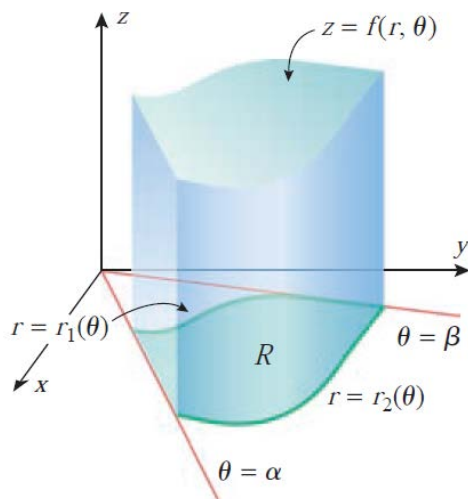
▲ Figure 14.3.3

A *polar rectangle* is a simple polar region for which the bounding polar curves are circular arcs. For example, Figure 14.3.3 shows the polar rectangle R given by

$$1.5 \leq r \leq 2, \quad \frac{\pi}{6} \leq \theta \leq \frac{\pi}{4}$$

DOUBLE INTEGRALS IN POLAR COORDINATES

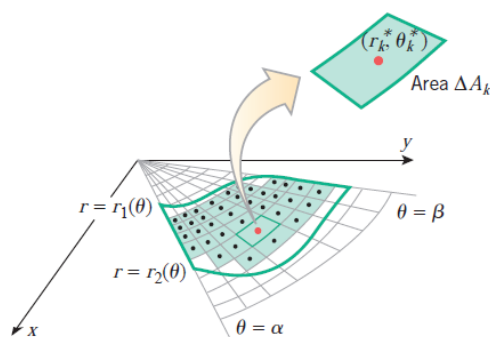
14.3.2 THE VOLUME PROBLEM IN POLAR COORDINATES Given a function $f(r, \theta)$ that is continuous and nonnegative on a simple polar region R , find the volume of the solid that is enclosed between the region R and the surface whose equation in cylindrical coordinates is $z = f(r, \theta)$ (Figure 14.3.4).



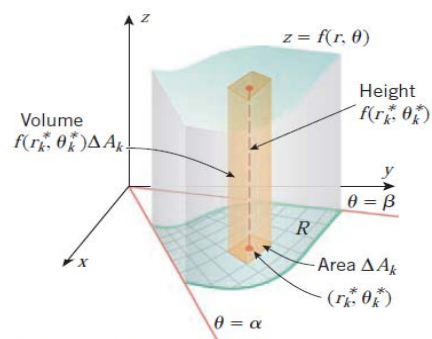
▲ Figure 14.3.4

If we now increase the number of subdivisions in such a way that the dimensions of the polar rectangles approach zero, then it seems plausible that the errors in the approximations approach zero, and the exact volume of the solid is

$$V = \lim_{n \rightarrow +\infty} \sum_{k=1}^n f(r_k^*, \theta_k^*) \Delta A_k \quad (1)$$



▲ Figure 14.3.5



▲ Figure 14.3.6

If $f(r, \theta)$ is continuous on R and has both positive and negative values, then the limit

$$\lim_{n \rightarrow +\infty} \sum_{k=1}^n f(r_k^*, \theta_k^*) \Delta A_k \quad (2)$$

represents the net signed volume between the region R and the surface $z = f(r, \theta)$ (as with double integrals in rectangular coordinates). The sums in (2) are called **polar Riemann sums**, and the limit of the polar Riemann sums is denoted by

$$\iint_R f(r, \theta) dA = \lim_{n \rightarrow +\infty} \sum_{k=1}^n f(r_k^*, \theta_k^*) \Delta A_k \quad (3)$$

which is called the **polar double integral** of $f(r, \theta)$ over R . If $f(r, \theta)$ is continuous and nonnegative on R , then the volume formula (1) can be expressed as

$$V = \iint_R f(r, \theta) dA \quad (4)$$

$$dA = r dr d\theta$$

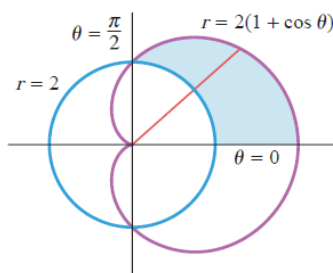
14.3.3 THEOREM If R is a simple polar region whose boundaries are the rays $\theta = \alpha$ and $\theta = \beta$ and the curves $r = r_1(\theta)$ and $r = r_2(\theta)$ shown in Figure 14.3.8, and if $f(r, \theta)$ is continuous on R , then

$$\iint_R f(r, \theta) dA = \int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r, \theta) r dr d\theta \quad (7)$$

► **Example 1** Evaluate

$$\iint_R \sin \theta dA$$

where R is the region in the first quadrant that is outside the circle $r = 2$ and inside the cardioid $r = 2(1 + \cos \theta)$.

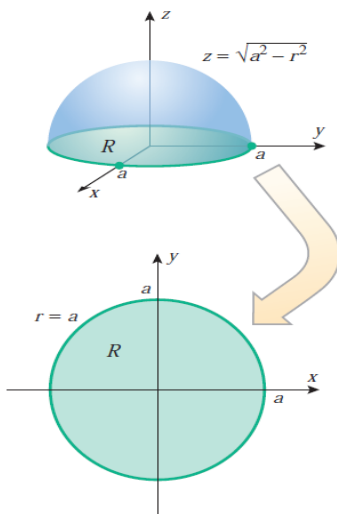


▲ Figure 14.3.10

Solution. The region R is sketched in Figure 14.3.10. Following the two steps outlined above we obtain

$$\begin{aligned}
 \iint_R \sin \theta \, dA &= \int_0^{\pi/2} \int_2^{2(1+\cos \theta)} (\sin \theta) r \, dr \, d\theta \\
 &= \int_0^{\pi/2} \left[\frac{1}{2} r^2 \sin \theta \right]_{r=2}^{2(1+\cos \theta)} d\theta \\
 &= 2 \int_0^{\pi/2} [(1 + \cos \theta)^2 \sin \theta - \sin \theta] d\theta \\
 &= 2 \left[-\frac{1}{3} (1 + \cos \theta)^3 + \cos \theta \right]_0^{\pi/2} \\
 &= 2 \left[-\frac{1}{3} - \left(-\frac{5}{3} \right) \right] = \frac{8}{3} \quad \blacktriangleleft
 \end{aligned}$$

► **Example 2** The sphere of radius a centered at the origin is expressed in rectangular coordinates as $x^2 + y^2 + z^2 = a^2$, and hence its equation in cylindrical coordinates is $r^2 + z^2 = a^2$. Use this equation and a polar double integral to find the volume of the sphere.



▲ Figure 14.3.11

Solution. In cylindrical coordinates the upper hemisphere is given by the equation

$$z = \sqrt{a^2 - r^2}$$

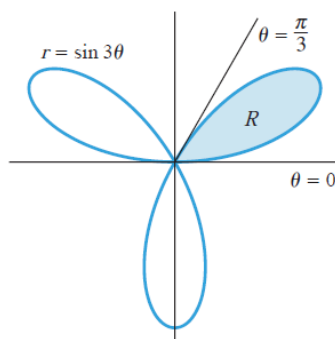
so the volume enclosed by the entire sphere is

$$V = 2 \iint_R \sqrt{a^2 - r^2} \, dA$$

where R is the circular region shown in Figure 14.3.11. Thus,

$$\begin{aligned} V &= 2 \iint_R \sqrt{a^2 - r^2} \, dA = \int_0^{2\pi} \int_0^a \sqrt{a^2 - r^2} (2r) \, dr \, d\theta \\ &= \int_0^{2\pi} \left[-\frac{2}{3} (a^2 - r^2)^{3/2} \right]_{r=0}^a d\theta = \int_0^{2\pi} \frac{2}{3} a^3 \, d\theta \\ &= \left[\frac{2}{3} a^3 \theta \right]_0^{2\pi} = \frac{4}{3} \pi a^3 \quad \blacktriangleleft \end{aligned}$$

► **Example 3** Use a polar double integral to find the area enclosed by the three-petaled rose $r = \sin 3\theta$.



▲ Figure 14.3.12

Solution. The rose is sketched in Figure 14.3.12. We will use Formula (8) to calculate the area of the petal R in the first quadrant and multiply by three.

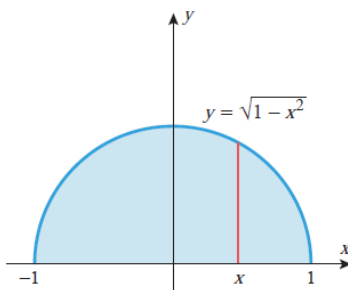
$$\begin{aligned}
 A &= 3 \iint_R dA = 3 \int_0^{\pi/3} \int_0^{\sin 3\theta} r \, dr \, d\theta \\
 &= \frac{3}{2} \int_0^{\pi/3} \sin^2 3\theta \, d\theta = \frac{3}{4} \int_0^{\pi/3} (1 - \cos 6\theta) \, d\theta \\
 &= \frac{3}{4} \left[\theta - \frac{\sin 6\theta}{6} \right]_0^{\pi/3} = \frac{1}{4}\pi \quad \blacktriangleleft
 \end{aligned}$$

■ CONVERTING DOUBLE INTEGRALS FROM RECTANGULAR TO POLAR COORDINATES

Sometimes a double integral that is difficult to evaluate in rectangular coordinates can be evaluated more easily in polar coordinates by making the substitution $x = r \cos \theta$, $y = r \sin \theta$ and expressing the region of integration in polar form; that is, we rewrite the double integral in rectangular coordinates as

$$\iint_R f(x, y) \, dA = \iint_R f(r \cos \theta, r \sin \theta) \, dA = \iint_{\text{appropriate limits}} f(r \cos \theta, r \sin \theta) r \, dr \, d\theta \quad (9)$$

► **Example 4** Use polar coordinates to evaluate $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{3/2} \, dy \, dx$.



▲ Figure 14.3.13

$$\begin{aligned}
 \int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{3/2} \, dy \, dx &= \iint_R (x^2 + y^2)^{3/2} \, dA \\
 &= \int_0^{\pi} \int_0^1 (r^3) r \, dr \, d\theta = \int_0^{\pi} \frac{1}{5} \, d\theta = \frac{\pi}{5} \quad \blacktriangleleft
 \end{aligned}$$

Example 1 Evaluate the following integrals by converting them into polar coordinates.

(a) $\iint_D 2xy \, dA$, D is the portion of the region between the circles of radius 2 and radius 5 centered at the origin that lies in the first quadrant.

(b) $\iint_D e^{x^2+y^2} \, dA$, D is the unit disk centered at the origin.

Solution

(a) $\iint_D 2xy \, dA$, D is the portion of the region between the circles of radius 2 and radius 5 centered at the origin that lies in the first quadrant.

First let's get D in terms of polar coordinates. The circle of radius 2 is given by $r = 2$ and the circle of radius 5 is given by $r = 5$. We want the region between the two circles, so we will have the following inequality for r .

$$2 \leq r \leq 5$$

Also, since we only want the portion that is in the first quadrant we get the following range of θ 's.

$$0 \leq \theta \leq \frac{\pi}{2}$$

Now that we've got these we can do the integral.

$$\begin{aligned} \iint_D 2xy \, dA &= \int_0^{\frac{\pi}{2}} \int_2^5 2(r \cos \theta)(r \sin \theta) r \, dr \, d\theta \\ \iint_D 2xy \, dA &= \int_0^{\frac{\pi}{2}} \int_2^5 r^3 \sin(2\theta) \, dr \, d\theta \\ &= \int_0^{\frac{\pi}{2}} \left. \frac{1}{4} r^4 \sin(2\theta) \right|_2^5 \, d\theta \\ &= \int_0^{\frac{\pi}{2}} \frac{609}{4} \sin(2\theta) \, d\theta \\ &= -\frac{609}{8} \cos(2\theta) \Big|_0^{\frac{\pi}{2}} \\ &= \frac{609}{4} \end{aligned}$$

(b) $\iint_D e^{x^2+y^2} dA$, D is the unit disk centered at the origin.

In this case we can't do this integral in terms of Cartesian coordinates. We will however be able to do it in polar coordinates. First, the region D is defined by,

$$0 \leq \theta \leq 2\pi$$

$$0 \leq r \leq 1$$

In terms of polar coordinates the integral is then,

$$\iint_D e^{x^2+y^2} dA = \int_0^{2\pi} \int_0^1 r e^{r^2} dr d\theta$$

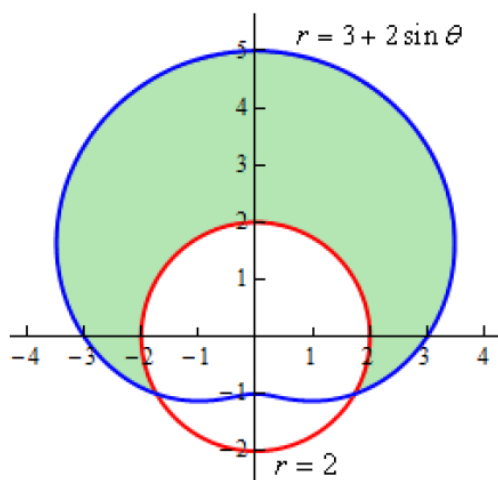
Notice that the addition of the r gives us an integral that we can now do. Here is the work for this integral.

$$\begin{aligned} \iint_D e^{x^2+y^2} dA &= \int_0^{2\pi} \int_0^1 r e^{r^2} dr d\theta \\ &= \int_0^{2\pi} \left. \frac{1}{2} e^{r^2} \right|_0^1 d\theta \\ &= \int_0^{2\pi} \frac{1}{2} (e-1) d\theta \\ &= \pi(e-1) \end{aligned}$$

Example 2 Determine the area of the region that lies inside $r = 3 + 2 \sin \theta$ and outside $r = 2$.

Solution

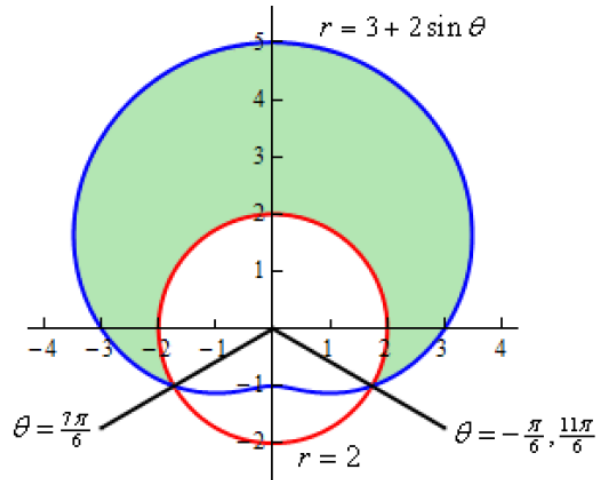
Here is a sketch of the region, D , that we want to determine the area of.



$$3 + 2 \sin \theta = 2$$

$$\sin \theta = -\frac{1}{2} \quad \Rightarrow \quad \theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

Here is a sketch of the figure with these angles added.



So, here are the ranges that will define the region.

$$-\frac{\pi}{6} \leq \theta \leq \frac{7\pi}{6}$$

$$2 \leq r \leq 3 + 2 \sin \theta$$

$$\begin{aligned} A &= \iint_D dA \\ &= \int_{-\pi/6}^{7\pi/6} \int_2^{3+2\sin\theta} r \, dr \, d\theta \\ &= \int_{-\pi/6}^{7\pi/6} \frac{1}{2} r^2 \Big|_2^{3+2\sin\theta} d\theta \\ &= \int_{-\pi/6}^{7\pi/6} \frac{5}{2} + 6 \sin \theta + 2 \sin^2 \theta \, d\theta \\ &= \int_{-\pi/6}^{7\pi/6} \frac{7}{2} + 6 \sin \theta - \cos(2\theta) \, d\theta \\ &= \left(\frac{7}{2} \theta - 6 \cos \theta - \frac{1}{2} \sin(2\theta) \right) \Big|_{-\pi/6}^{7\pi/6} \\ &= \frac{11\sqrt{3}}{2} + \frac{14\pi}{3} = 24.187 \end{aligned}$$

Practice Problem:

Chapter-14.3:

Problems: 7-10, 21-26, 27-34

Thank you for your attention!